

応用数学 2章 第5回 Basic 解答

【Basic】

問題 106

次の微分方程式を解け. (1) $\frac{dx}{dt} - x = e^{2t}$, $x(0) = 1$
(2) $\frac{dx}{dt} + x = 1$, $x(0) = 0$

解答:

$$\begin{aligned} (1) \quad sX - 1 - X &= \frac{1}{s-2} \quad \text{より} \quad (s-1)X = 1 + \frac{1}{s-2} = \frac{s-1}{s-2} \\ X &= \frac{1}{s-2} \quad \text{より} \quad x(t) = e^{2t} \\ (2) \quad sX + X &= \frac{1}{s} \quad \text{より} \quad X = \frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1} \\ x(t) &= 1 - e^{-t} \end{aligned}$$

問題 107

次の微分方程式を解け. (1) $\frac{d^2x}{dt^2} - \frac{dx}{dt} - 2x = e^t$ ($t = 0$ のとき $x = 0$, $\frac{dx}{dt} = 0$) (2) $\frac{d^2x}{dt^2} - 2\frac{dx}{dt} + 5x = 0$ ($t = 0$ のとき $x = 0$, $\frac{dx}{dt} = 1$)

解答:

$$\begin{aligned} (1) \quad s^2X - sX - 2X &= \frac{1}{s-1}, \quad (s^2 - s - 2)X = \frac{1}{s-1} \\ X &= \frac{1}{(s-1)(s-2)(s+1)} = -\frac{1}{2} \cdot \frac{1}{s-1} + \frac{1}{3} \cdot \frac{1}{s-2} + \frac{1}{6} \cdot \frac{1}{s+1} \\ x(t) &= \frac{1}{6}e^{-t} - \frac{1}{2}e^t + \frac{1}{3}e^{2t} \\ (2) \quad (s^2 - 2s + 5)X &= 1, \quad X = \frac{1}{(s-1)^2 + 4} \\ x(t) &= \frac{1}{2}e^t \sin 2t \end{aligned}$$

問題 108

次の微分方程式を解け. (1) $\frac{d^2x}{dt^2} - 2\frac{dx}{dt} = 0$, $x(0) = 1$, $x(1) = e^2$ (2) $\frac{d^2x}{dt^2} + 9x = 3$, $x(0) = 1$, $x\left(\frac{\pi}{6}\right) = 0$

解答:

$$\begin{aligned} (1) \quad (s^2 - 2s)X &= s - 2 \quad (x(0) = 1, \quad x'(0) = c \text{ とする}) \\ x(1) = e^2 \text{ の条件から } c &= 2. \quad X = \frac{s + c - 2}{s(s-2)} \\ c = 2 \text{ のとき } X &= \frac{1}{s-2} \quad \text{より} \quad x(t) = e^{2t} \\ (2) \quad x(0) = 1, \quad x(\pi/6) &= 0 \text{ の条件で解くと } (s^2 + 9)X = \frac{3}{s + c + \frac{3}{s}} \end{aligned}$$

$$\begin{aligned} x(t) &= \cos 3t + \frac{c}{3} \sin 3t + \frac{1}{3}(1 - \cos 3t) \quad x(\pi/6) = 0 \\ \text{より } c &= -1. \quad x(t) = \frac{1}{3} + \frac{2}{3} \cos 3t - \frac{1}{3} \sin 3t \end{aligned}$$

問題 109

次の微分方程式の一般解を求めよ. (1) $\frac{dx}{dt} + 4x = 1$
(2) $\frac{d^2x}{dt^2} - 9x = 0$ (3) $\frac{d^2x}{dt^2} + 16x = t$

解答:

$$\begin{aligned} (1) \quad x(0) &= c_1 \text{ として } (s+4)X = c_1 + \frac{1}{s} \\ X &= \frac{c_1}{s+4} + \frac{1}{s(s+4)} \quad \text{より} \quad x(t) = c_1 e^{-4t} + \frac{1}{4}(1 - e^{-4t}) \\ \text{一般解: } x(t) &= C e^{-4t} + \frac{1}{4} \\ (2) \quad (s^2 - 9)X &= s c_1 + c_2 \\ \text{一般解: } x(t) &= C_1 e^{3t} + C_2 e^{-3t} \\ (3) \quad (s^2 + 16)X &= s c_1 + c_2 + \frac{1}{s^2} \\ \text{一般解: } x(t) &= C_1 \cos 4t + C_2 \sin 4t + \frac{t}{16} \end{aligned}$$